Arduino data conversion – 1023 or 1024?

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Voltage at the analog input pins can be measured by using the analogRead function. Since it returns the 10-bit integer of the ADC output, the value falls in the range from 0 to 1023. Looking at the examples on the Internet, you can find two ways how the voltage is calculated:

1. analogRead(A0)*(5.0/1023.0);
2. analogRead(A0)*(5.0/1024.0);

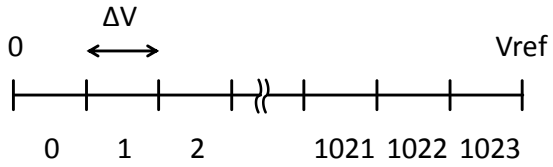
They are not the same, so which is the correct one?

The “ADC Conversion Result” section in the datasheet of the microcontroller says [1]:

ADC = Vin/Vref*1024.

Why 1024? You may think that if Vref is the highest voltage and 1023 is the highest code, then 1023 must be the digital equivalent of Vref. Not really!

The input voltage range is 0 to Vref and this range is divided into 1024 uniform intervals, since 1024 different code values are possible. Therefore, each code is associated with an interval of voltage, which has a width of ΔV = Vref/1024.



For example, ADC code of 0 means that 0 ≤ Vin < ΔV; 1 means that ΔV ≤ Vin< 2·ΔV, etc.

The largest code, 1023 means that 1023·ΔV ≤ Vin< 1024·ΔV, in other words, Vref-ΔV ≤ Vin< Vref. The full range of 0 ≤ Vin< Vref is covered, but Vref is not included! You can also find a maximum reading of 1.999 V on digital voltmeters used in the 2V range. Successive approximation ADCs halve the inspected range in each step, therefore Vref can only be divided by power of two [2].

So, 1024 is the right value, it is the digital equivalent of Vref as the datasheet states.

One may say that it does not matter, which one you use, since the difference is negligible. Indeed, it is below 0.1%, but there is no reason to use an incorrect value. If you get used to leaving the correct ways, you may run into problems unexpectedly.

Digitization is very common and there are lower resolutions when using the correct value is particularly important. The datasheet [1] says: “*If a lower resolution than 10 bits is needed, the input clock frequency to the ADC can be higher than 200kHz to get a higher sample rate”*. Using the 8 most significant bits the maximum reading is 255. Vref/255 is not four times larger than Vref/1023, while Vref/256 do equal 4*Vref/1024.

In a day there are 24 hours. If you look at the hours displayed on a digital clock (on your microwave oven, notebook, smart phone, smart watch, etc.), you can see numbers from 0 to 23, a digitized value of the current time of the day. Each hour represents 1/24th part of the whole day. You know that 23 does not correspond to the last moment of the day, right?

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References

1. Atmega328 datasheet, https://www.microchip.com/wwwproducts/en/ATmega328P
2. The Data Conversion Handbook, Edited by Walt Kester, Newnes, 2005, ISBN 0-7506-7841-0. https://www.analog.com/en/education/education-library/data-conversion-handbook.html, See the Coding and Quantizing chapter, https://www.analog.com/media/en/training-seminars/design-handbooks/Data-Conversion-Handbook/Chapter2.pdf .